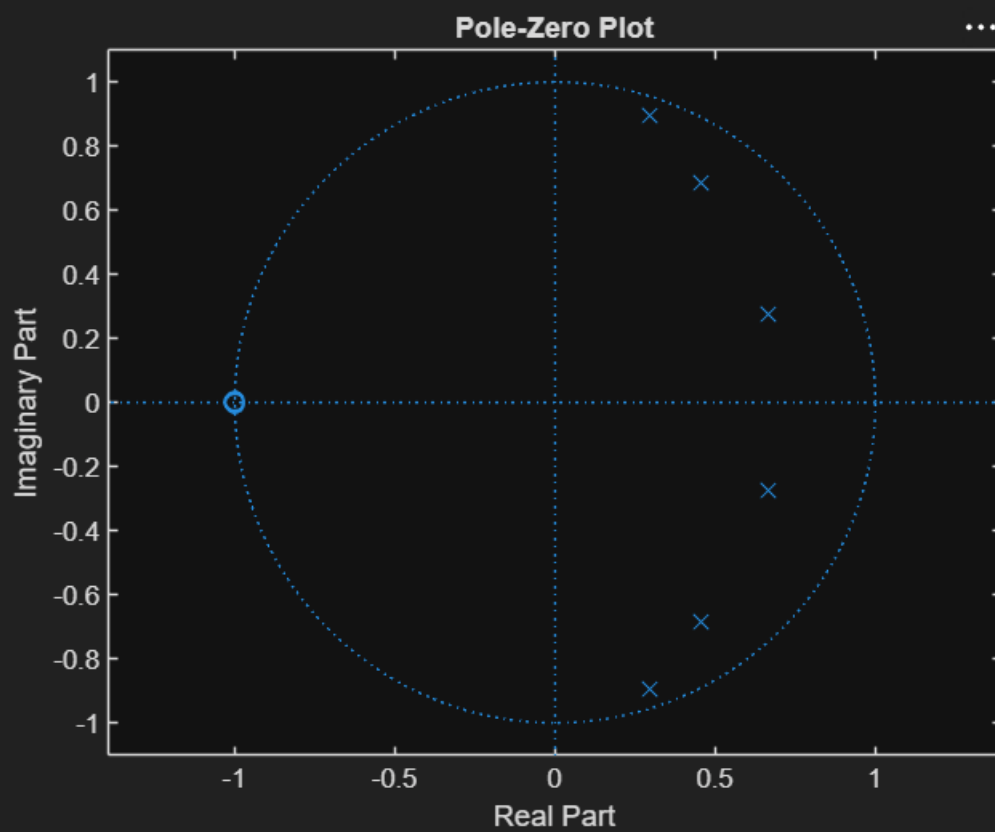
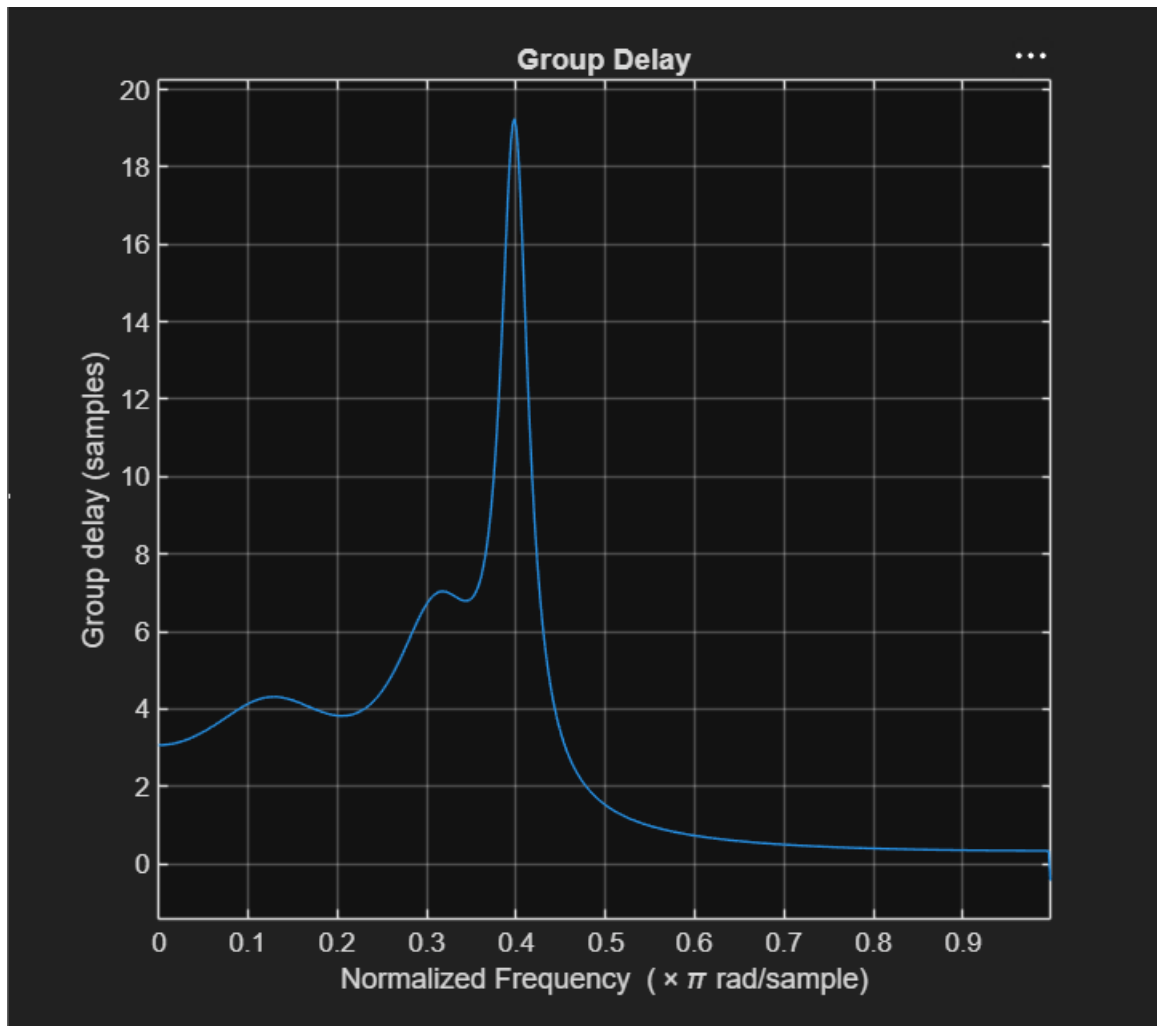
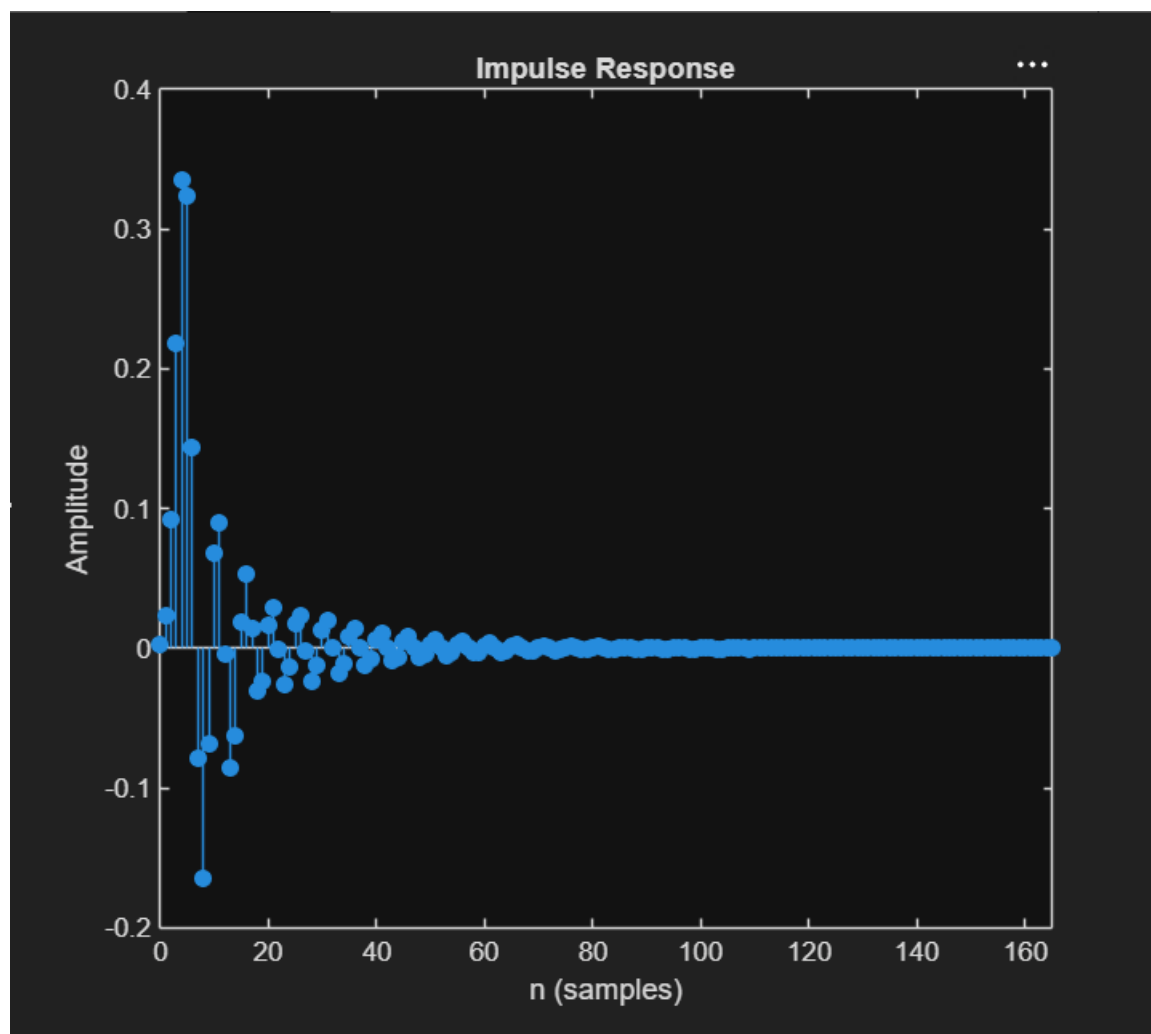
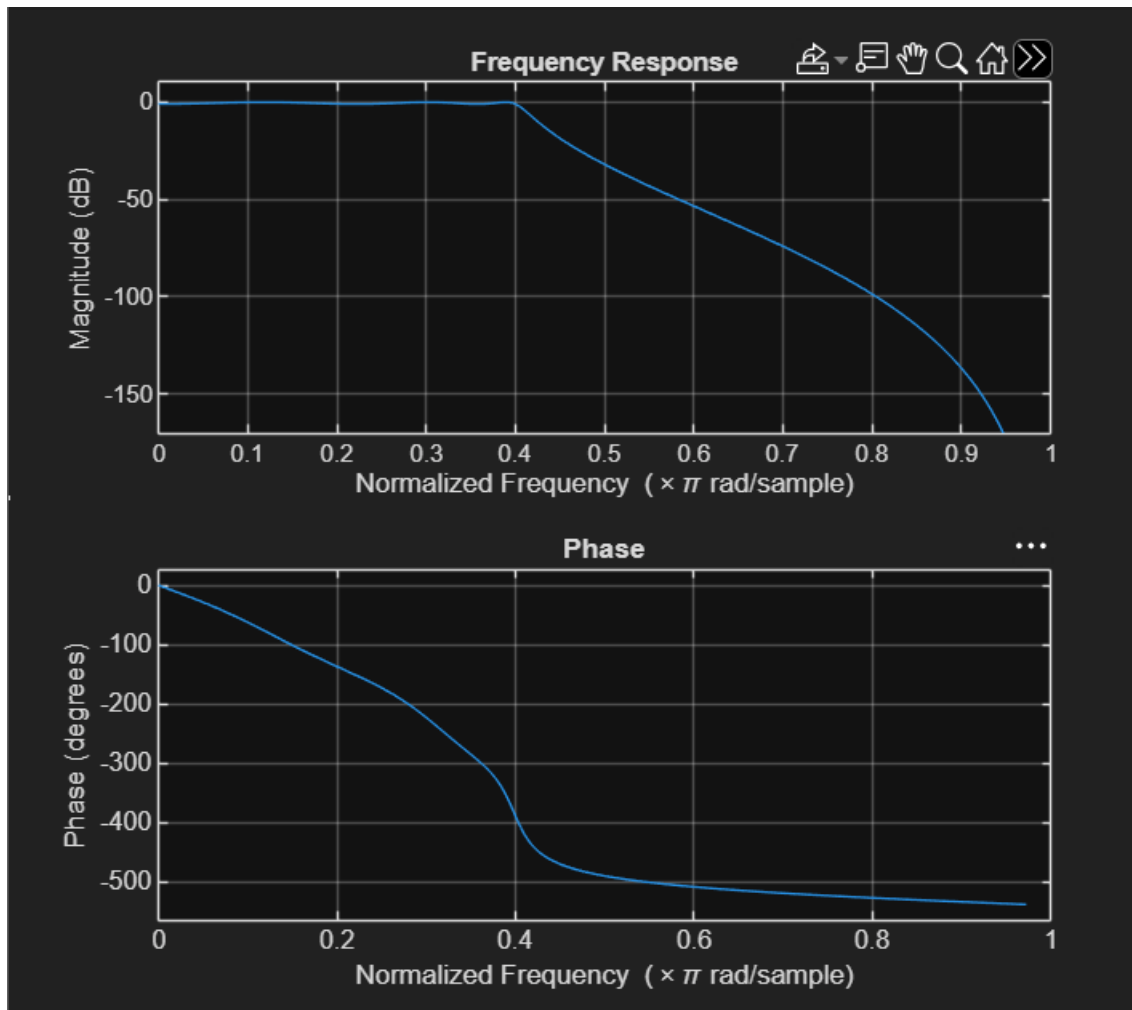










PROGRAM

```
clc;
```

```
clear;
```

```
close all;
```

```
% EX NO: 2
```

```
% Chebyshev IIR Filter
```

```
% DATE:
```

```
% 2.1 Design and Performance Evaluation of a
```

```
% Chebyshev Type-I IIR Low-Pass Filter using MATLAB
```

```
% Filter Specifications
```

```

Wp = 0.4;    % Passband Edge (Normalized)
Ws = 0.55;   % Stopband Edge (Normalized)
Rp = 1;      % Passband Ripple (dB)
As = 40;     % Stopband Attenuation (dB)

% Calculate Minimum Filter Order
[N, Wn] = cheb1ord(Wp, Ws, Rp, As);

% Design Chebyshev Type-I Low-Pass Filter
[b, a] = cheby1(N, Rp, Wn, 'low');

% Display Filter Details
fprintf('Filter Order = %d\n', N);
fprintf('Normalized Cutoff Frequency = %.4f\n', Wn);

% Stability Check
if all(abs(roots(a)) < 1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% -----
% Frequency Response
% -----

figure;
freqz(b, a);
title('Frequency Response');

```

```
% -----  
  
% Impulse Response  
  
% -----  
  
figure;  
impz(b, a);  
title('Impulse Response');
```

```
% -----  
  
% Group Delay  
  
% -----  
  
figure;  
grpdelay(b, a);  
title('Group Delay');
```

```
% -----  
  
% Pole-Zero Plot  
  
% -----  
  
figure;  
zplane(b, a);  
title('Pole-Zero Plot');
```